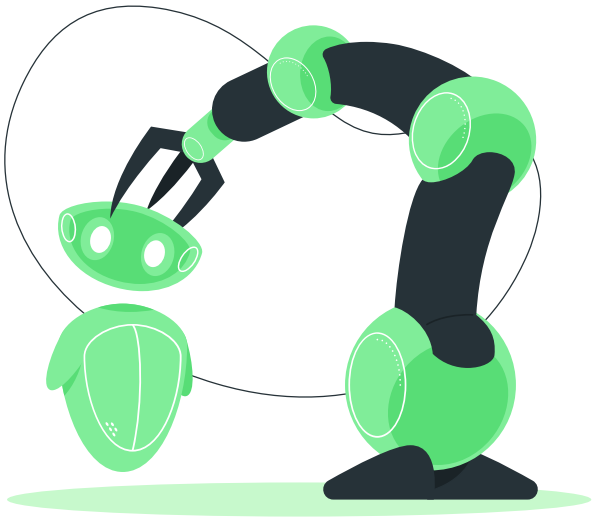
Robotics
CS 3015

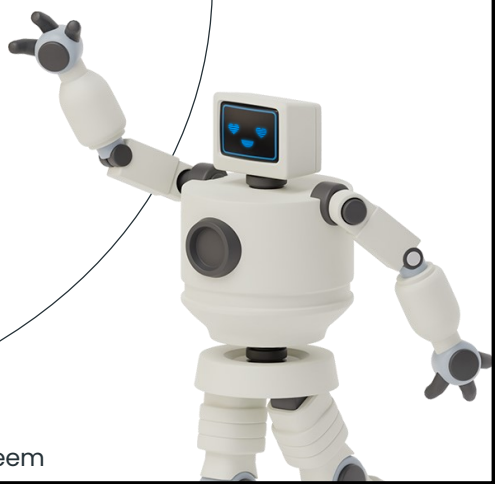
Umarah Qaseem



1

3

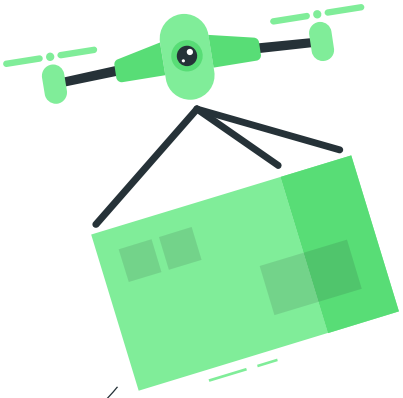
- 3D coordinate Frame
- 3D Point and Vector
- 3D coordinate Frame
- Pose
- Relative Pose
- Applications in Real World
- Rotation
- Transformation



2

What did we learn?

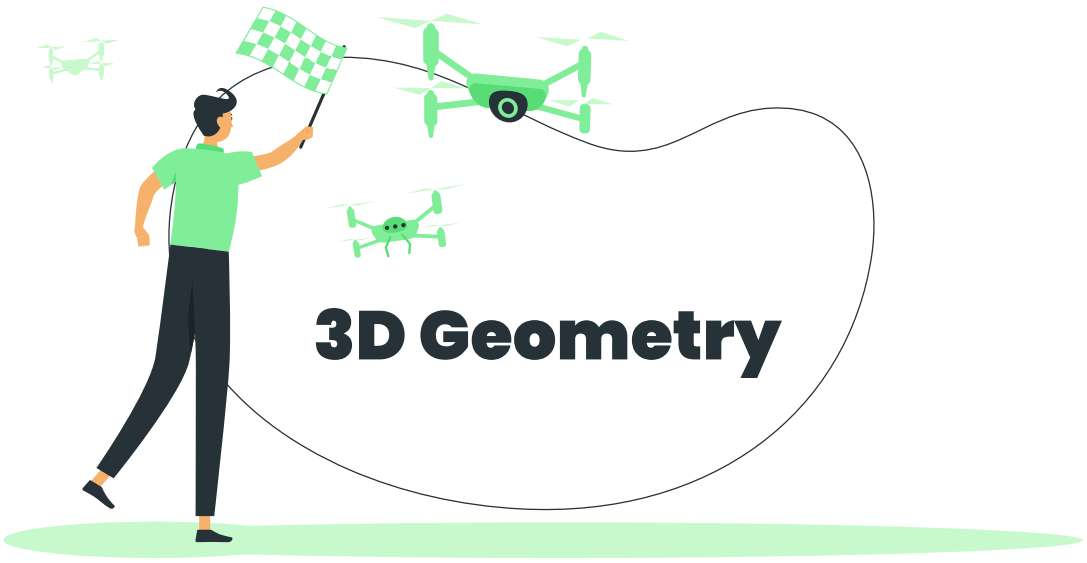
?



Revision

Robotics – Umarah Qaseem

3



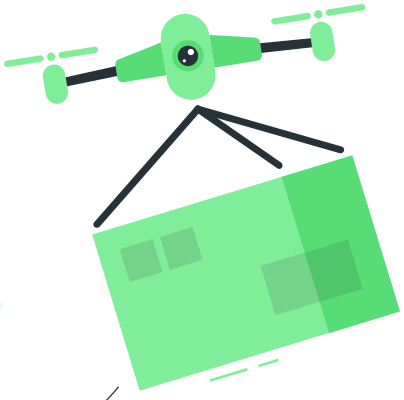
3D Geometry

Robotics – Umarah Qaseem

5

Need in Robotics?

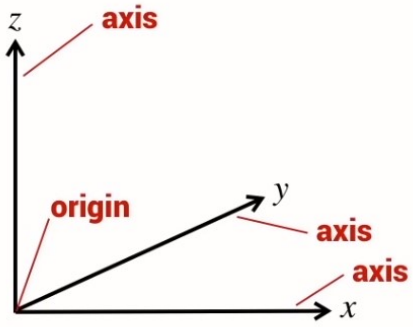
3D Geometry



Robotics – Umarah Qaseem

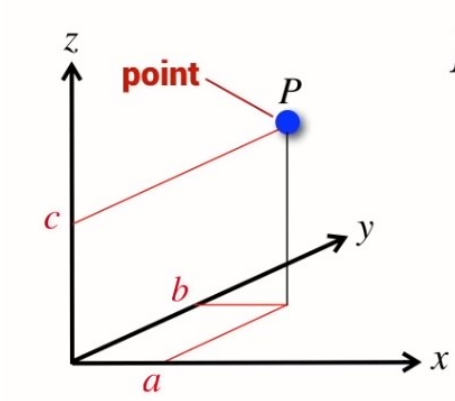
6

3D coordinate Frame



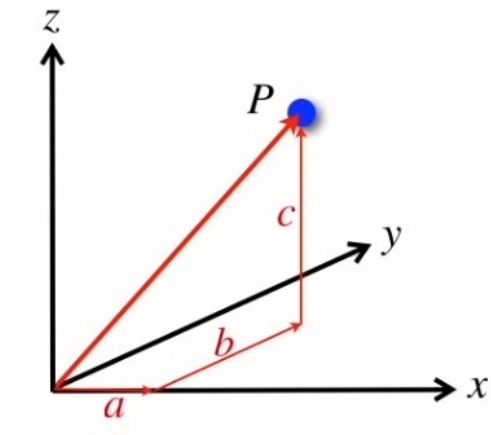
7

Point in 3D



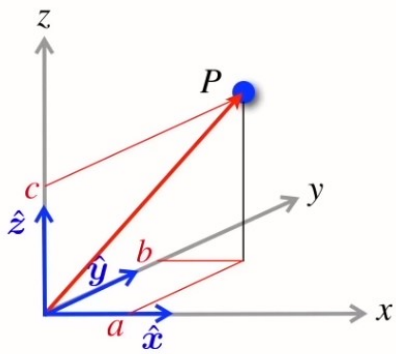
$$P = (a, b, c)$$

Vector in 3D



$$\boldsymbol{p} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

In terms of orthogonal unit vectors

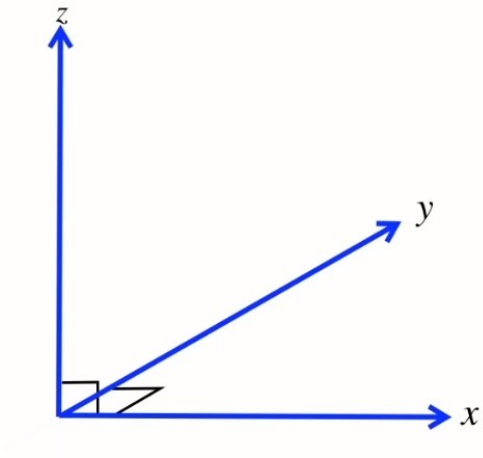


$$\boldsymbol{p} = a\hat{\boldsymbol{x}} + b\hat{\boldsymbol{y}} + c\hat{\boldsymbol{z}}$$

Robotics – Umarah Qaseem

10

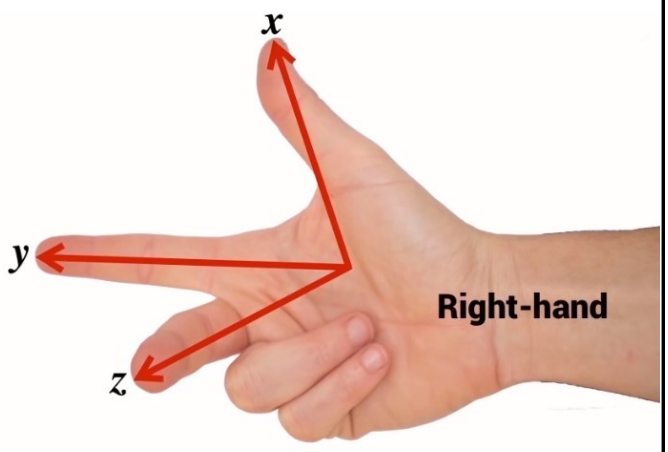
3D Coordinate Frame



Robotics – Umarah Qaseem

12

Right Hand Rule



Robotics – Umarah Qaseem

14

Pose in 3D

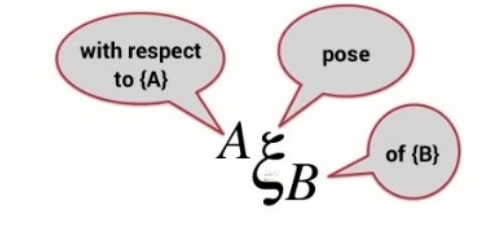
Mathematics and derivation

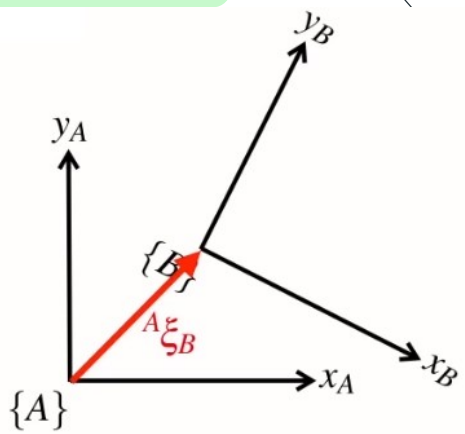


Robotics – Umarah Qaseem

15

Recap 2D Pose



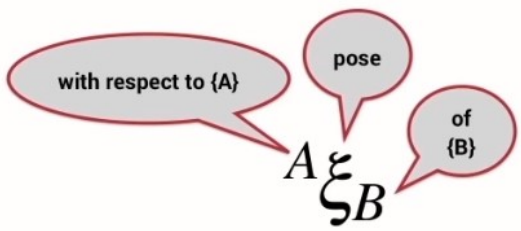


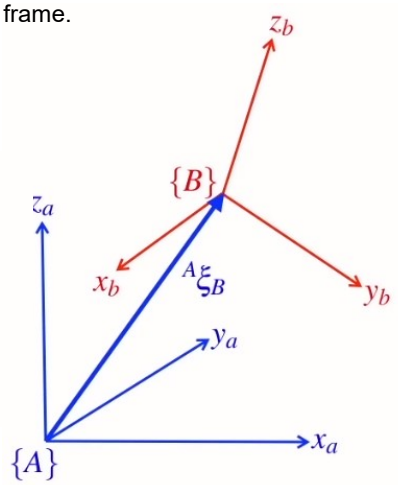
Revision

16

3D Pose

Pose is described with respect to another coordinate frame.
Two components are?

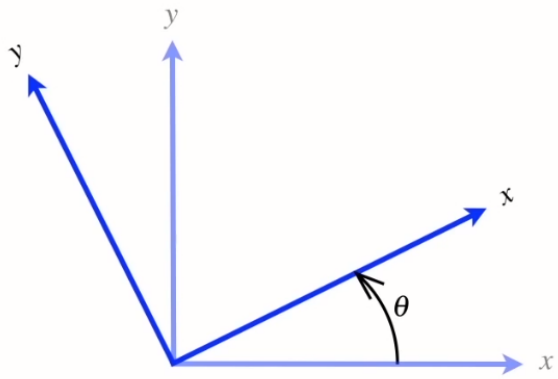




17

Where is the Z axis?

- This is a positive rotation around Z axis!!



Robotics – Umarah Qaseem

19

Application In Real World

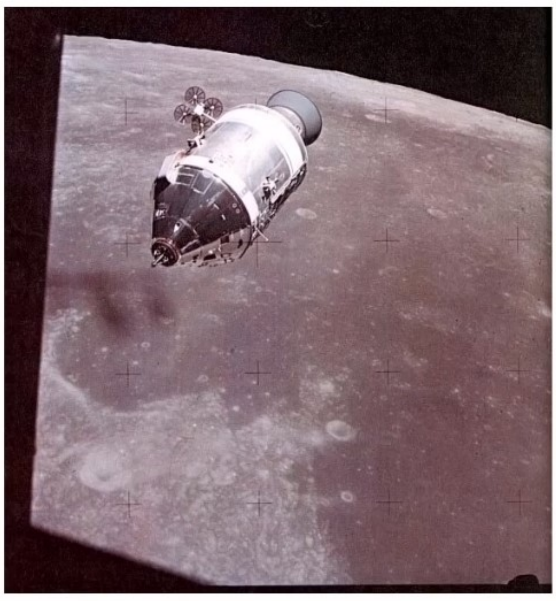


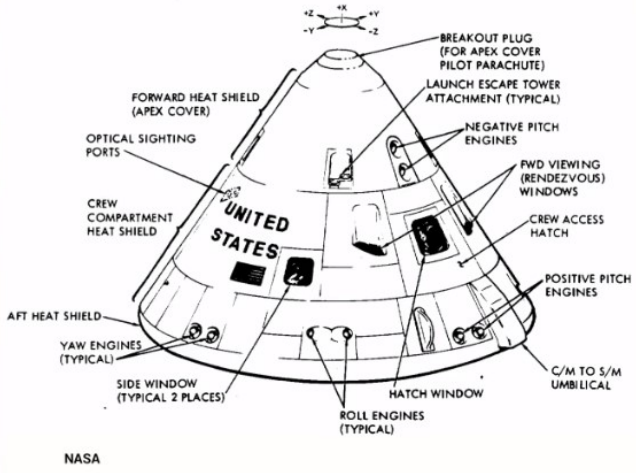
21

In Real World



22





SM2A-02-BLOCK II-(1) Apollo Operations Handbook
NASA/GSFC | Public domain

23

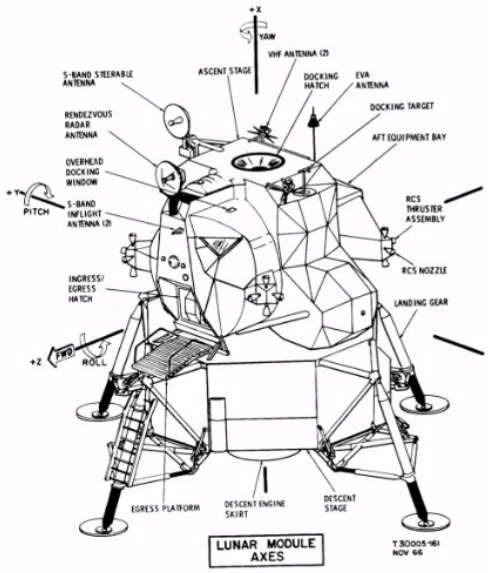


Image: NASA

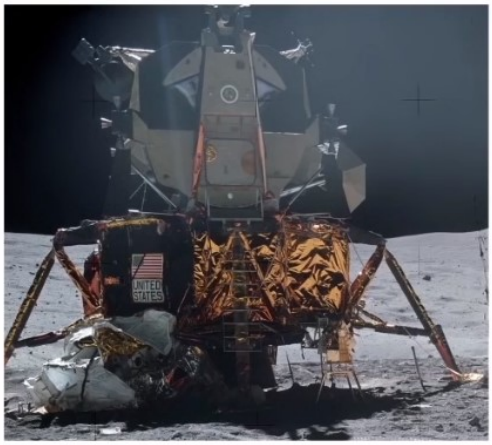
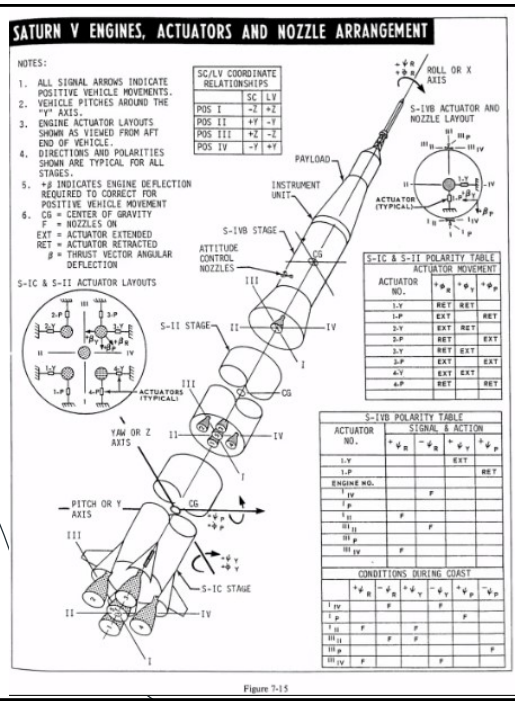


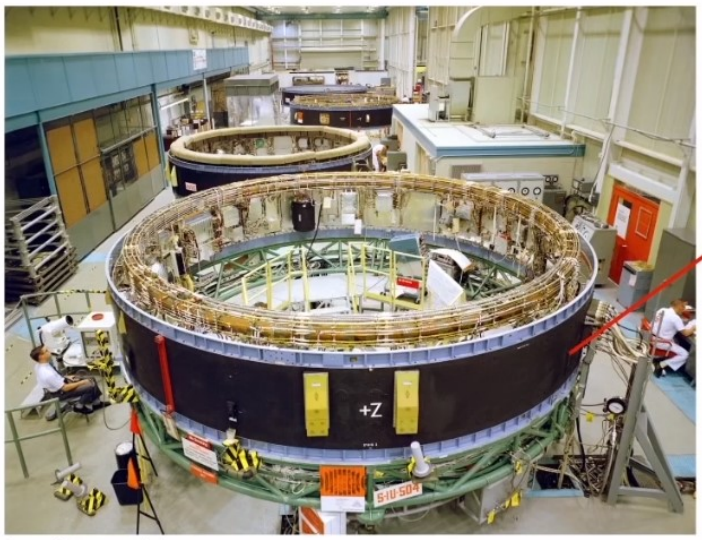
Image: NASA

24





25



Saturn V Instrument Unit
Image credit: NASA/MSFC



26

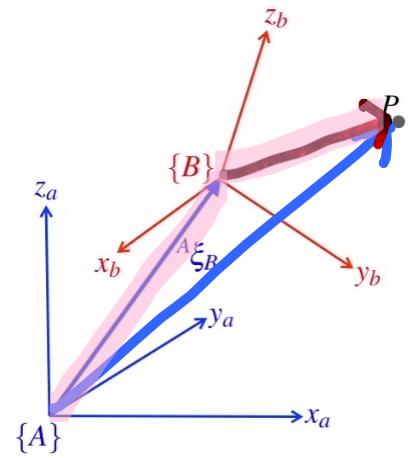
Relative Pose

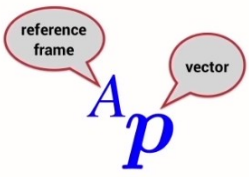
3D



27

Point and Vector

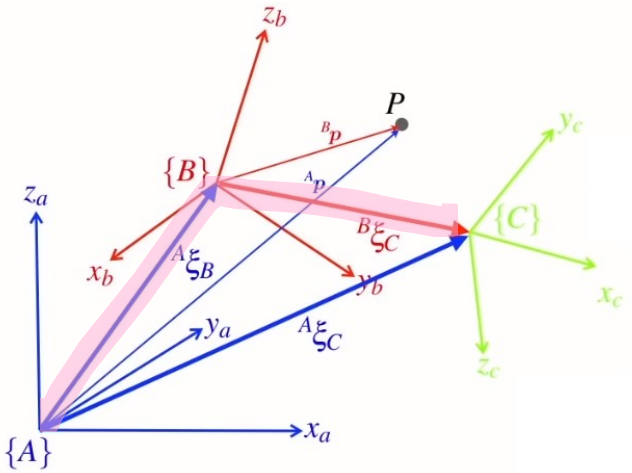




$${}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p}$$

transform vector from one frame to another

Relative Position



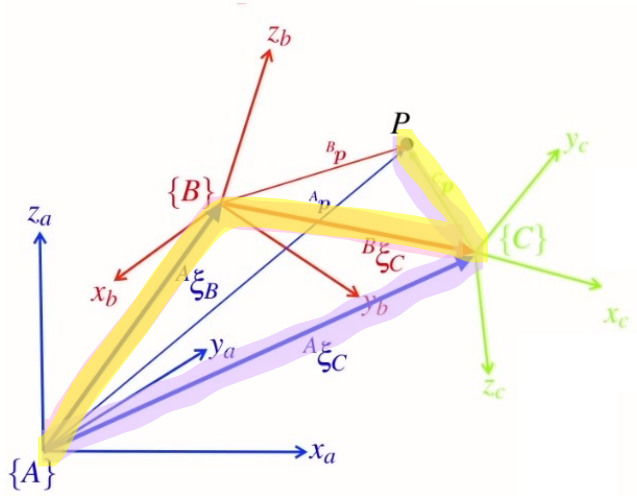
$${}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p}$$

transform vector from one frame to another

composition

$${}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

Relative Position



$$ {}^A p = {}^A\xi_B \cdot {}^B p $$

transform vector from one frame to another

composition

$$ {}^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C $$

$$ {}^A p = ({}^A\xi_B \oplus {}^B\xi_C) \cdot {}^C p $$

Relative Position

$$ {}^X\xi_Y \oplus {}^Y\xi_Z = {}^X\xi_Z $$

$$ {}^X p = {}^X\xi_Y \bullet {}^Y p $$

$$ \xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 $$

$$ \xi \oplus 0 = \xi $$

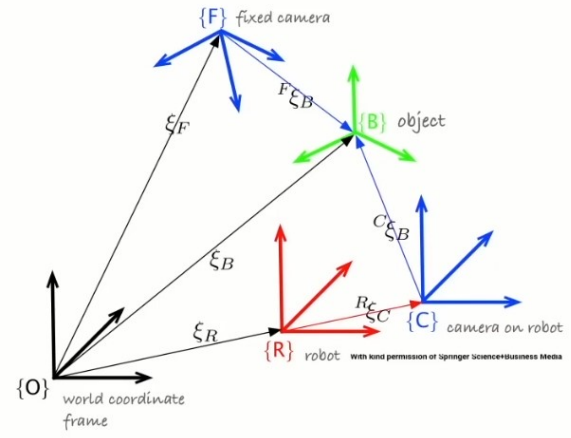
$$ \ominus {}^X\xi_Y = {}^Y\xi_X $$

$$ \xi \ominus \xi = 0 \qquad \ominus\xi \oplus \xi = 0 $$

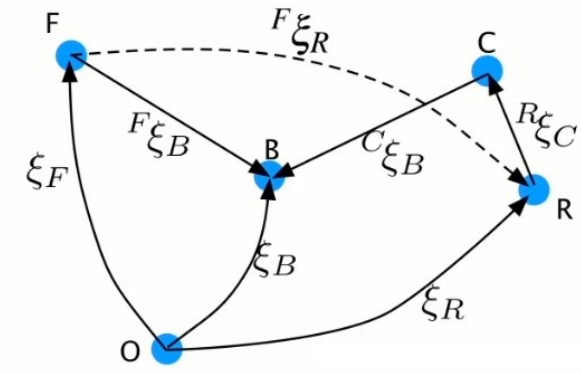
$$ \xi \ominus 0 = \xi $$

same rules as for 2D

Relative Position

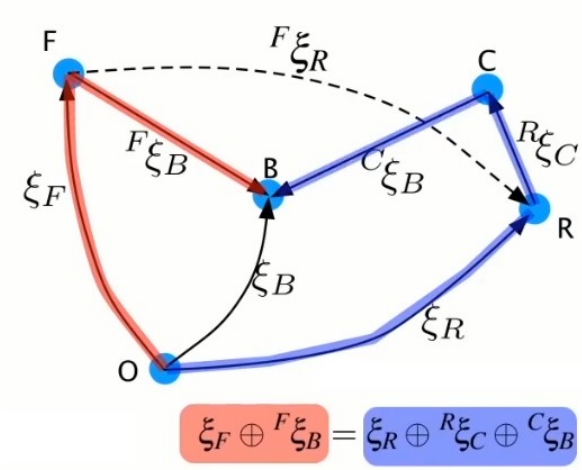


Relative Position



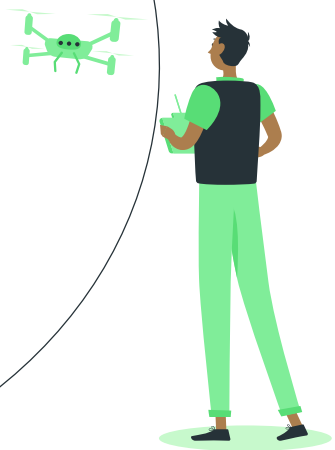
$$\xi_F \oplus {}^F\xi_B = \xi_R \oplus {}^R\xi_C \oplus {}^C\xi_B$$

Relative Position



Rotation in 3D

Mathematics and derivation

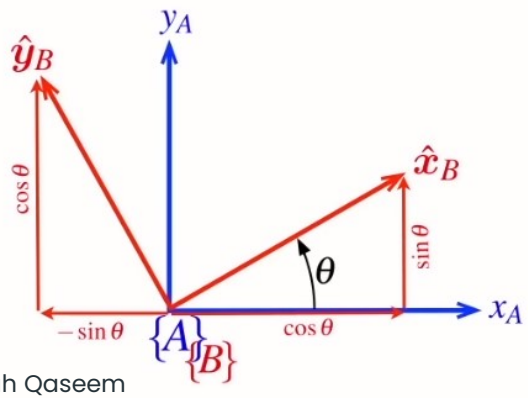


Recap Rotation in 2D

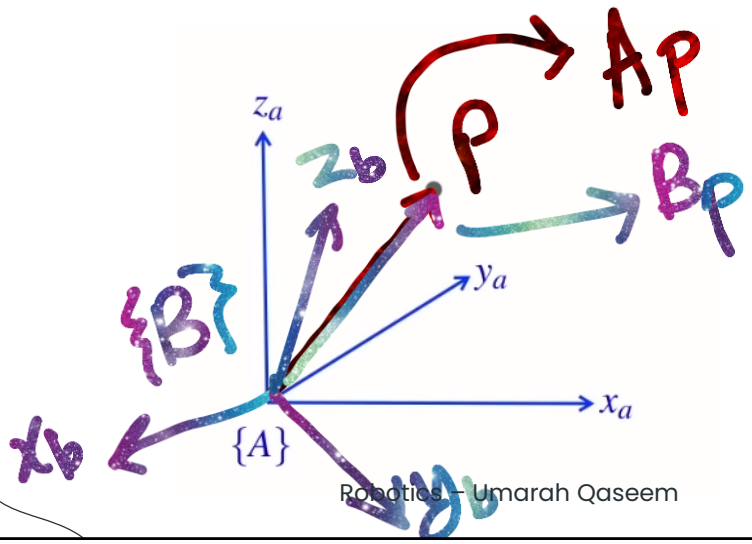
$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix}$$

new x-axis

new y-axis



Taking same scenario in 3D

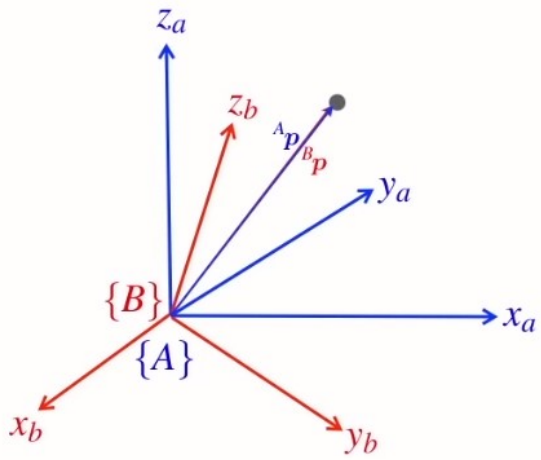


3D Rotation Matrix

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ {}^{A}z \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ {}^{B}z \end{pmatrix}$$

new x-axis | new y-axis | new z-axis

- Again, we are describing new axis B in terms of old old axis A.
- This matrix transforms vectors from new frame B to old frame A.



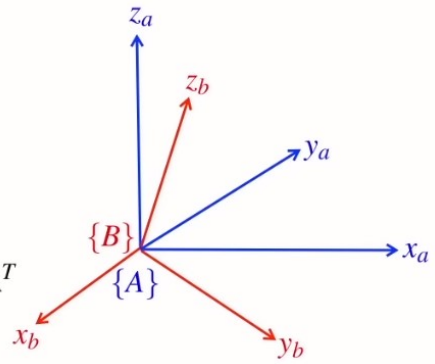
Properties of Rotation Matrix

- Same as 2D rotation matrix

Properties of Rotation Matrix

$$ {}^{A}\mathbf{R}_{B} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} $$

- An orthogonal (orthonormal) matrix
- Each column is a **unit** length **vector**
- Each column is **orthogonal** to all other columns
- The inverse is the same as the transpose $\mathbf{R}^{-1} = \mathbf{R}^{T}$
- The determinant is +1 $det(\mathbf{R}) = 1$
  - ➡ the length of a vector is unchanged by rotation



40

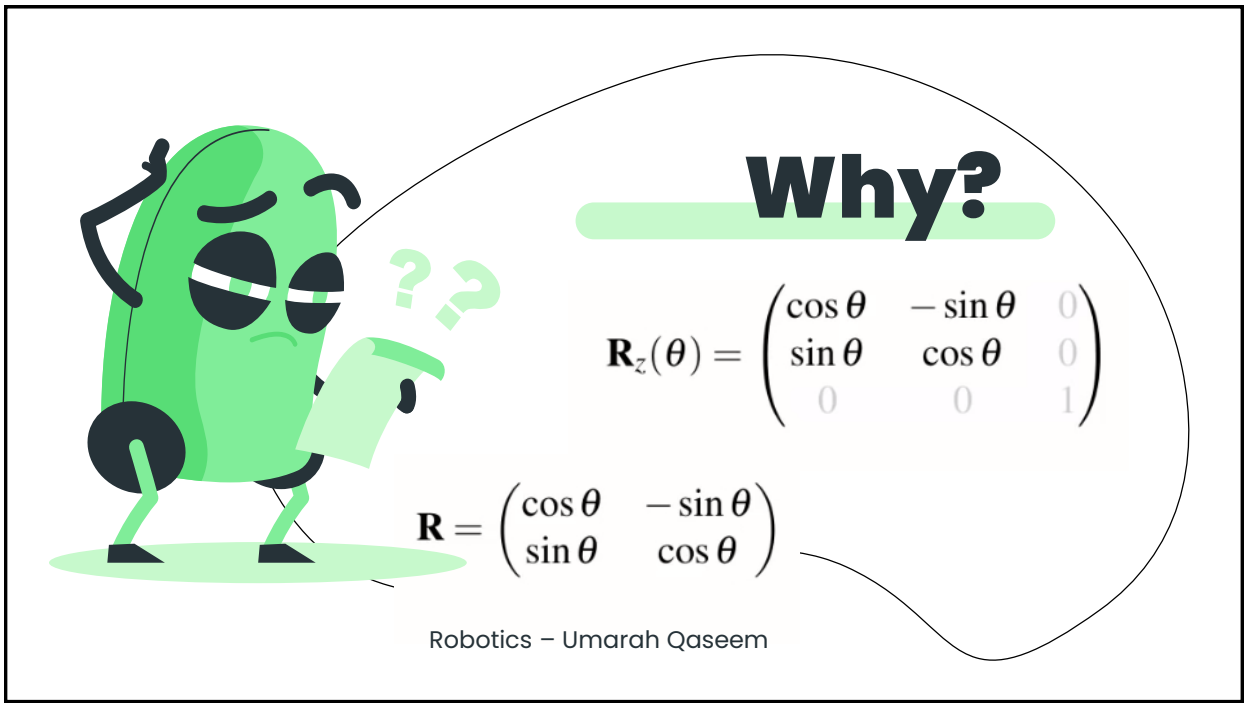
Elementary Rotation Matrices

$$ \mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} $$

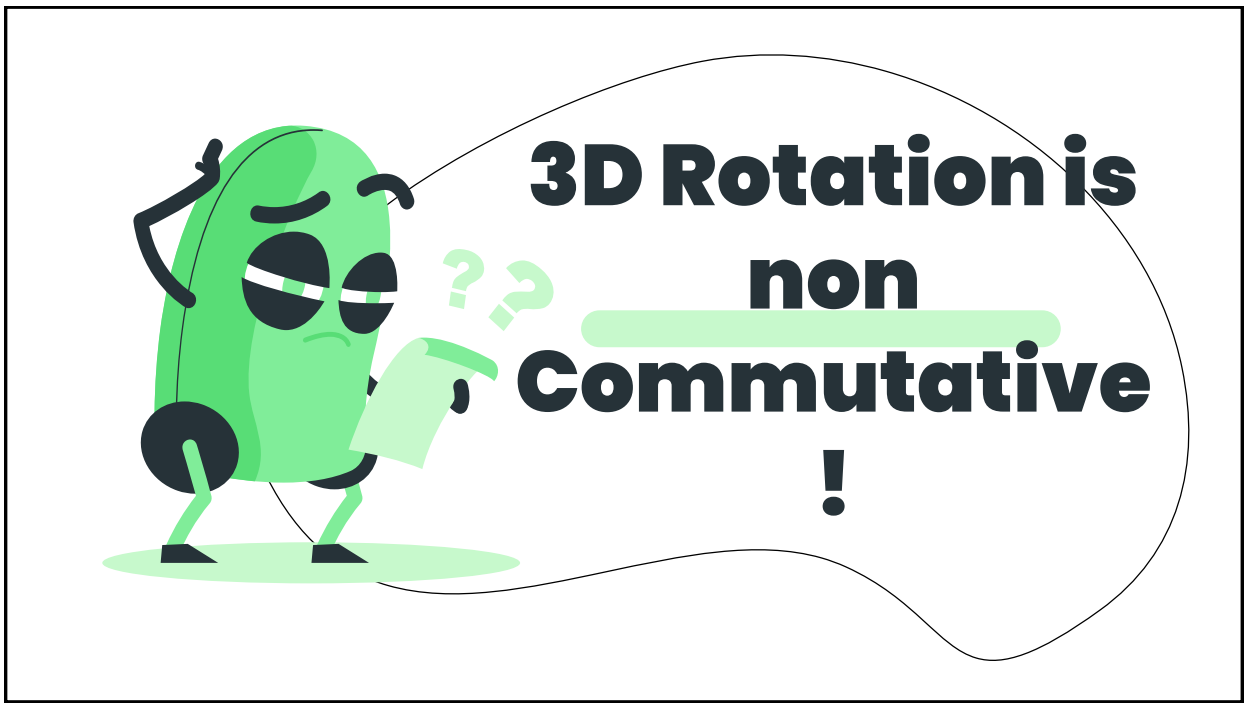
$$ \mathbf{R}_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} $$

$$ \mathbf{R}_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} $$

41


Why?
$\mathbf{R}_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$
$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
Robotics – Umarah Qaseem

42


3D Rotation is non Commutative !

44

Transformation in 3D

Mathematics and derivation

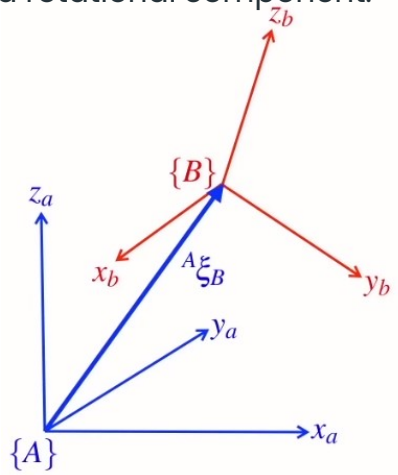


45

To Represent Pose

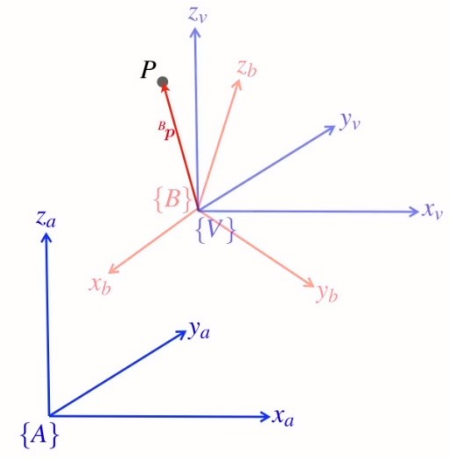
- Pose has a translational component + a rotational component.

- Vector + Euler Angles
- Vector + roll, pitch, yaw angles
- Vector + Quaternion
- Homogeneous Transformations

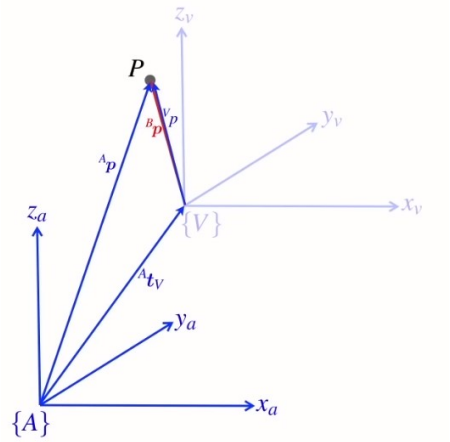


46

Translational Component



Translational Component



Derivation

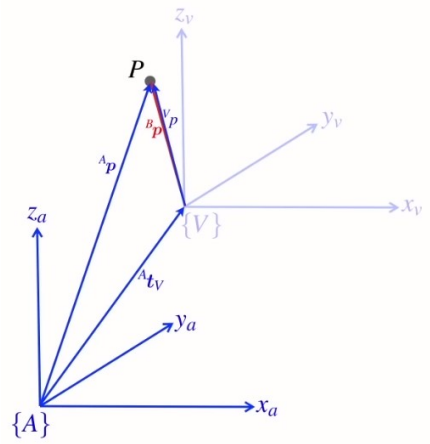
$$^{V}\boldsymbol{p} = {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\boldsymbol{p} \qquad \{V\} \parallel \{A\}$$

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{V} + {}^{V}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{B} + {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

Reason?



Derivation

$$^{A}\boldsymbol{p} = {}^{A}\boldsymbol{t}_{B} + {}^{A}\mathbf{R}_{B}\,{}^{B}\boldsymbol{p}$$

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ {}^{A}z \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ {}^{B}z \end{pmatrix} + \begin{pmatrix} {}^{A}x_B \\ {}^{A}y_B \\ {}^{A}z_B \end{pmatrix}$$

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ {}^{A}z \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & {}^{A}x_B \\ r_{21} & r_{22} & r_{23} & {}^{A}y_B \\ r_{31} & r_{32} & r_{33} & {}^{A}z_B \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ {}^{B}z \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ {}^{A}z \\ 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} & {}^{A}x_B \\ r_{21} & r_{22} & r_{23} & {}^{A}y_B \\ r_{31} & r_{32} & r_{33} & {}^{A}z_B \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ {}^{B}z \\ 1 \end{pmatrix}$$

$$^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\boldsymbol{t}_{B} \\ 0,0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}}$$

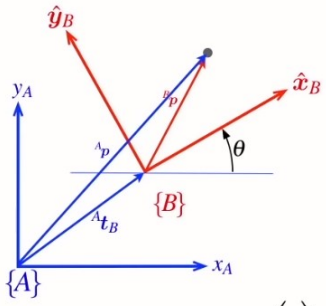
Properties of Homogeneous transformation matrix

- Describes relative pose as a 4x4 matrix

homogeneous vector | homogeneous transformation | homogeneous vector

$$ {}^{A}\tilde{\boldsymbol{p}} = \begin{pmatrix} {}^{A}\mathbf{R}_B & {}^{A}\boldsymbol{t}_B \\ 0,0,0 & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}} $$

$$ {}^{A}\tilde{\boldsymbol{p}} = {}^{A}\mathbf{T}_B \, {}^{B}\tilde{\boldsymbol{p}} \qquad {}^{A}\mathbf{T}_B = \begin{pmatrix} {}^{A}\mathbf{R}_B & {}^{A}\boldsymbol{t}_B \\ 0,0,0 & 1 \end{pmatrix} $$



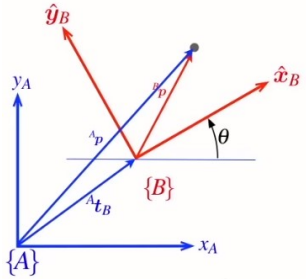
$$ \boldsymbol{p} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad \tilde{\boldsymbol{p}} = \begin{pmatrix} a \\ b \\ c \\ 1 \end{pmatrix} $$

Let's make things real

- Pose is just a Matrix!

$$ {}^{A}\xi_B \sim {}^{A}\mathbf{T}_B $$

- Composition is just a matrix product!



$$ {}^{X}\xi_Y \oplus {}^{Y}\xi_Z \mapsto {}^{X}\mathbf{T}_Y \, {}^{Y}\mathbf{T}_Z $$

- Negation is a matrix Inverse!

$$ \ominus\xi \mapsto \mathbf{T}^{-1} $$

- Inverse of T is NOT its transpose!
- Vector transformation is a matrix-vector product
- Remember dot operator?
- (which transforms vector from one Frame to another)

$$ \xi \cdot \boldsymbol{p} \mapsto \mathbf{T}\tilde{\boldsymbol{p}} $$

Thing to Remember!

Use lots of Coordinate Frames!

They are your friends!
Even NASA uses them!

53